Macroeconomics Assignment

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1 Consumption as a Fraction of GDP

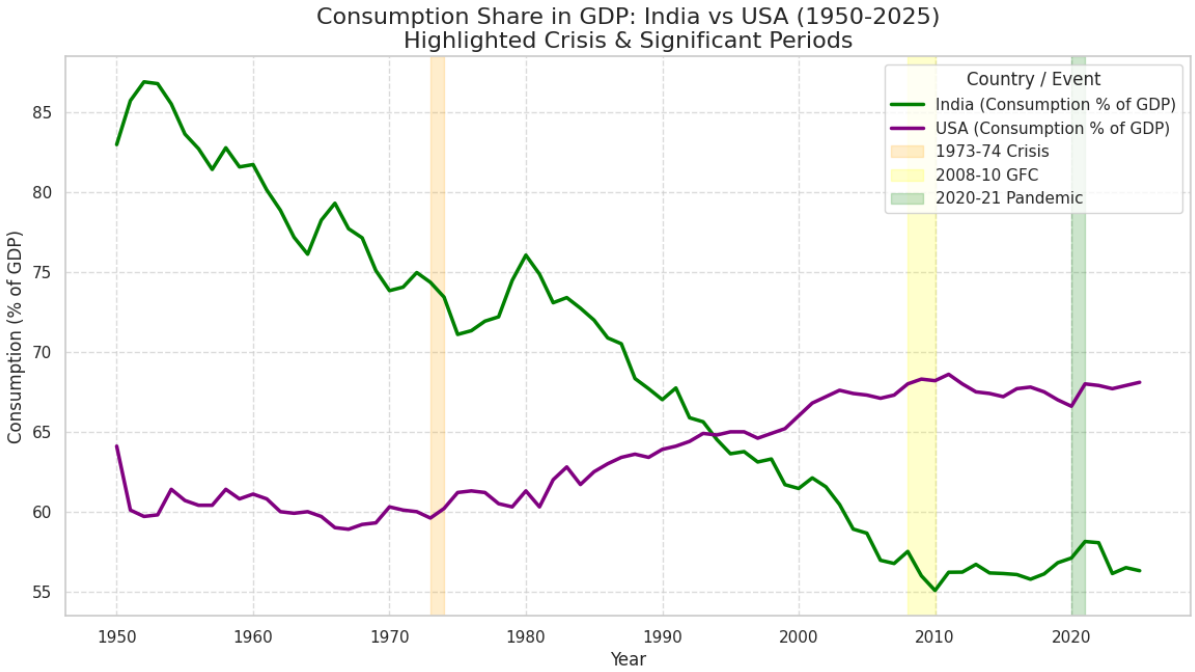

Figure 1: Consumption Share of GDP: India vs USA (1950–2025)

The consumption share of GDP exhibits a clear structural divergence between India and the United States over the long run. In India, consumption as a fraction of GDP declines steadily from very high levels in the early decades (around 80–85%) to nearly 55–58% in recent years.

In terms of volatility, consumption is more stable in the United States. People in the US have higher incomes. Strong social safety nets like unemployment benefits protect consumption during downturns. They have well developed credit markets, therefore households smooth their spending over time. Therefore they have stable consumption per gdp. In India, it's less stable because people have lower incomes majorly though the informal sector. There is limited access to credit due to which many people can't smoothen

their consumption when income changes. This makes consumption more sensitive to economic fluctuation.

During the 1973 oil crisis,In India, the sharp rise in oil prices increased import costs and inflation. This reduced people's real incomes, and since most households depended on current income and had limited access to credit or social protection, they had to cut down consumption. At the same time, government spending pressures and structural needs kept other components of GDP from falling as much, so the share of consumption in GDP declined. In the United States, although the oil shock also slowed the economy, households were better protected. Strong social safety nets and well-developed credit markets allowed people to maintain their consumption levels despite the slowdown. Meanwhile, investment fell more sharply due to uncertainty and higher costs. Because consumption did not fall as much as other components, its share in GDP actually increased.

During the 2008 global financial crisis, the United States experienced a noticeable increase in consumption share. The mechanism was driven by a collapse in investment. Due to the financial crisis there was a credit crunch which led to decrease in investment and therefore decrease in GDP, while consumption declined only moderately due to smoothing behavior. As a result, GDP fell more than consumption, leading to an increase in the consumption share (C/Y $\uparrow$). In contrast, India was less exposed to global financial markets, hence there was a smaller decline in investment, so the GDP was relatively stable. There was only a mild dip in consumption share followed by recovery.

The COVID-19 pandemic in 2020–21 created a sharp divergence between the two economies. In India, strict lockdowns and disruptions in the informal sector led to a severe income shock: economic activity halted therefore income and consumption decreased leading to decrease in GDP. The decline in GDP was more than that in Consumption and income resulting in an increase in the consumption share (C/Y $\uparrow$). In the United States, however, large fiscal stimulus measures supported household incomes: government transfers increased which maintained the disposable income and hence consumption, while GDP still declined. Consequently, the consumption share remained stable or even increased slightly (C/Y $\uparrow$).

2 Investment as a Fraction of GDP

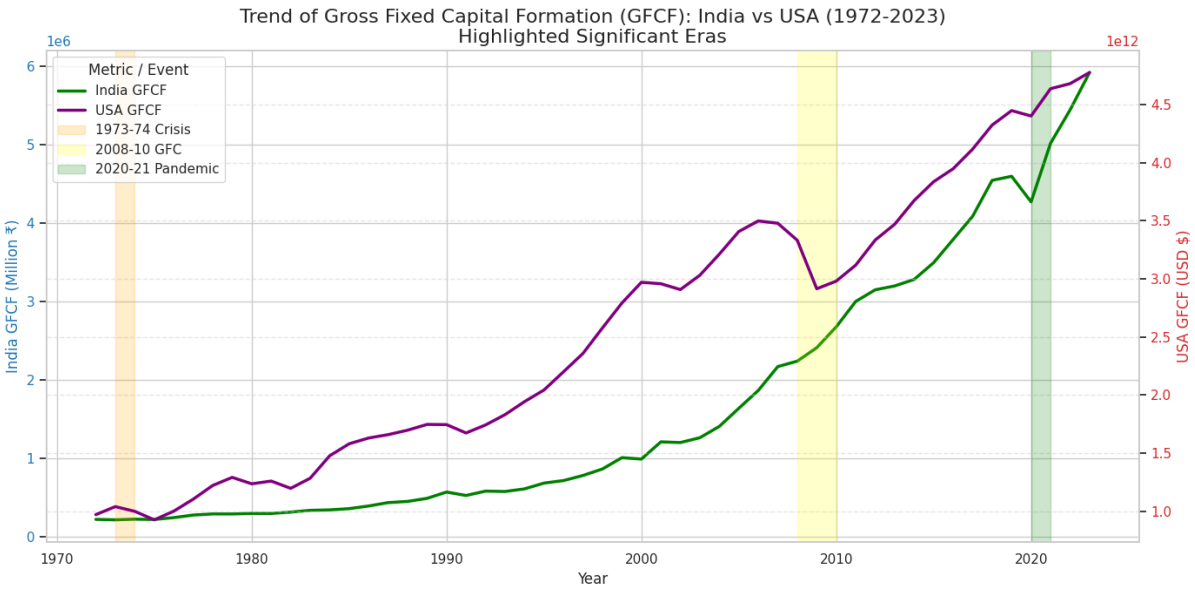

Figure 2: Investment Share of GDP (GFCF): India vs USA

Both the USA and India have rising trends in Investment as a share of GDP (GFCF/GDP). In the United States, the ratio remains relatively stable (especially until the 1990s) reflecting an economy with steady growth and developed financial markets. In India, the investment share is more variable, initially lower and fluctuating, but rising post-liberalization.

During the 1973 oil crisis, In India, the sharp rise in oil prices increased import costs and inflation which reduced profits and increased uncertainty. Firms cut down the investment therefore investment share in GDP decreased. In contrast, the United States experienced stagnation, as it had deep financial markets and firms could still access the credit so the investment didn't collapse. Therefore investment share remained relatively stable.

During the 2008 global financial crisis, India's exports decreased and there was demand uncertainty. The capital inflows decreased and there was an increase in financial constraints, therefore firms delayed or even cancelled the investment due to which its share in GDP fell down drastically. In the United States, although investment fell, GDP declined more sharply, leading to a temporary increase in the investment share.

During the COVID-19 pandemic in 2020, both countries saw an increase in investment share. Due to lockdown there was a sharp decline in consumption which declined GDP. This increased share of investment in GDP.

3 Government Spending as a Fraction of GDP

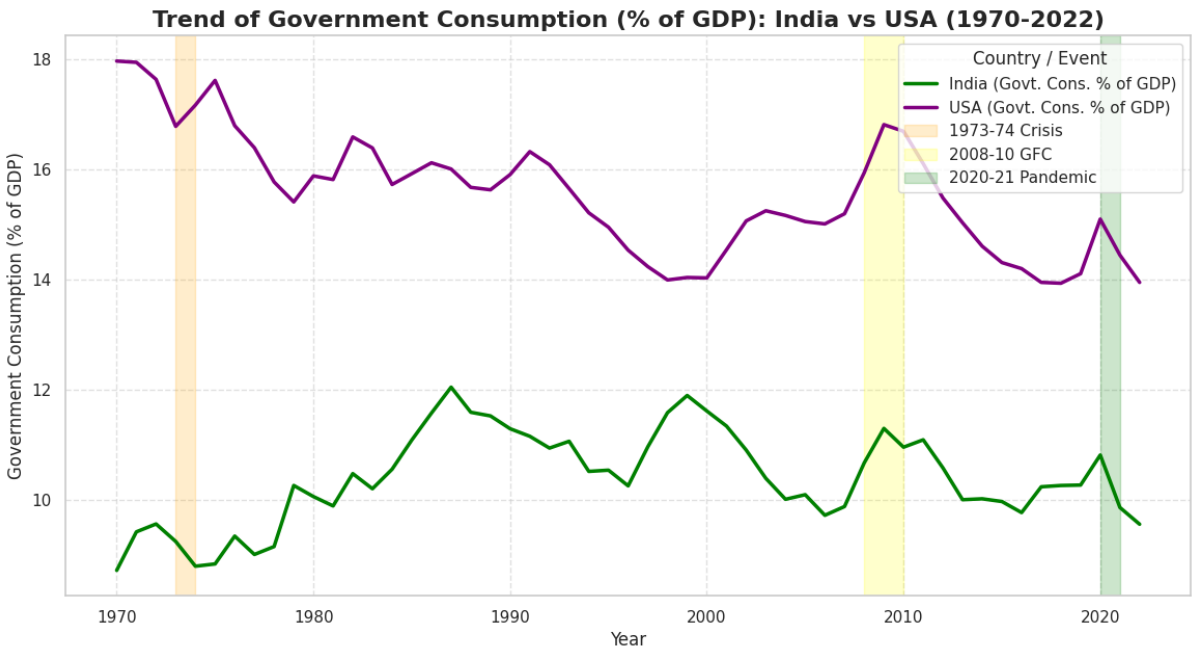

Figure 3: Government Consumption Share of GDP: India vs USA

Till 1990, India and the US showed opposite trends in government consumption as a share of GDP due to differences in their stage of economic development. Until 1990, there was a huge role of government with higher spending on administration, subsidies and development. There was an expansion in the public sector. Due to this Gov expenditure as share of GDP showed an increasing trend. On the other hand, during the Regan era (1980), the US government followed policies of tax cut and deregulation. It focused on reducing the role of the state and promoting private sector growth. Due to this government spending did not increase as fast as the gdp which led to a fall in share of government expenditure as share of GDP.

After 1990, the trends became similar in both countries. India adopted economic liberalisation in 1991, reducing the direct role of the government in production and allowing markets to play a larger role. This stabilised government consumption as a share of GDP. The US, already a mature economy, also exhibited relatively stable government consumption with only cyclical changes. As a result, both countries show similar, more stable trends after 1990.

In terms of volatility, both countries show sharp movements in the graph. This shows that in both countries, government consumption responds abruptly to shocks and hence is volatile.

During the 1973 oil crisis, in India, oil shock led to high inflation.The government

could not increase consumption spending proportionately, and GDP was also affected, so government consumption as a share of GDP declined. In the US, the government increased spending (including defense and public services) to support the economy during the slowdown. Since this spending rose relative to GDP, government consumption share increased.

During the 2008 global financial crisis, G/Y first increased and then decreased in both countries. In 2008-09: due to the financial crisis, GDP fell down sharply, and the government increased expenditure (stimulus, relief measures). Therefore, the government share in GDP increased. In 2009-10: Economic recovery began, GDP started rising again and stimulus measures were reduced which led to a fall in the share of government expenditure.

During the COVID-19 pandemic in 2020, there was a sharp contraction in GDP, but at the same time, a large part of government support was in the form of transfers (cash support, subsidies) rather than direct consumption spending. Since transfers are not counted in government consumption, the increase in total fiscal support did not fully translate into higher government consumption. In addition, lockdowns disrupted government service delivery (like administration, public services, etc.), which limited actual consumption expenditure. As the economy started recovering, GDP (denominator) increased faster, while government consumption did not rise proportionately.

4 Payments to Labor as a Share of GDP

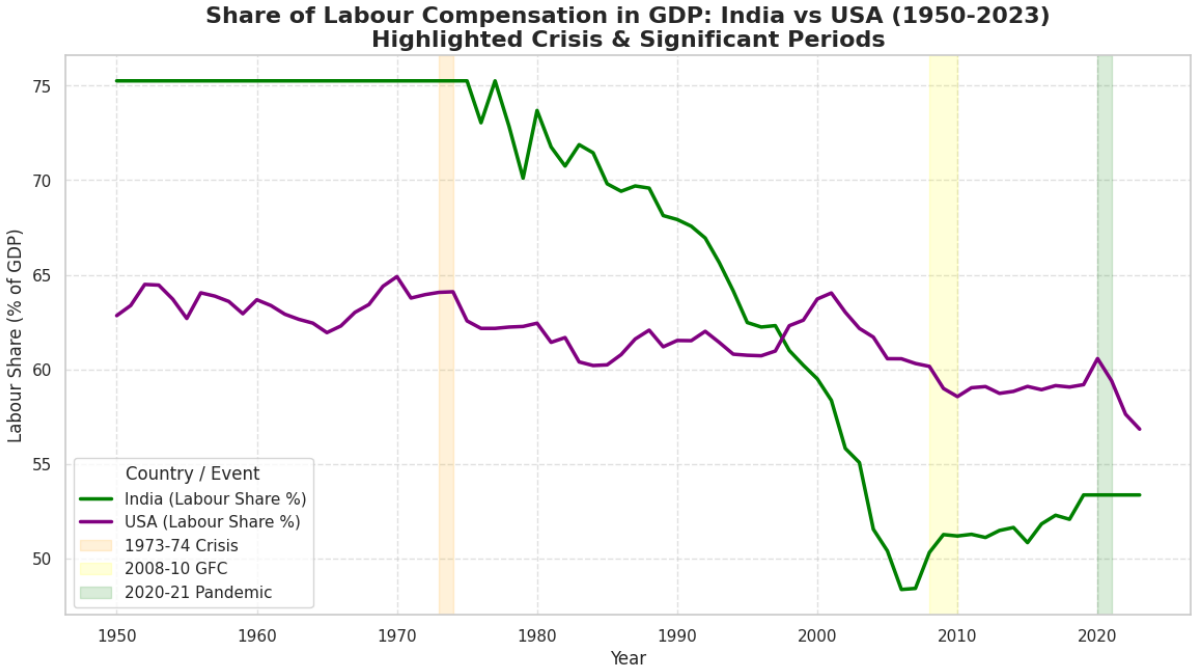

Figure 4: Labor Share of GDP: India vs USA

The share of labor compensation in GDP (labor share) reflects how income is distributed between workers and capital. In the United States, the labor share has been relatively stable over long periods, with a mild declining trend in recent decades. In India, the labor share has been constant till the 1970s and then decreased till the 2000s and then again there is an increasing trend.

Government consumption as a share of GDP is relatively stable compared to investment. The United States shows low volatility, with smooth and gradual changes due to stable fiscal policy. India exhibits higher volatility, with more noticeable fluctuations driven by changing fiscal needs and economic conditions.

During the 1973 oil crisis, the labor share remained relatively constant in both countries. This is because the shock raised production costs and reduced overall output, but both wages and profits adjusted simultaneously. As a result, the distribution between labor and capital did not change significantly, keeping the labor share stable.

During COVID-19 pandemic in 2020, labor share declined in India but increased in the US. In India, the shock disproportionately affected informal and low-wage workers, leading to sharp income and job losses, which reduced labor's share in GDP. In contrast, in the US, large fiscal support measures such as wage subsidies helped sustain household incomes, while corporate profits declined. This led to an increase in the labor share, as labor income was relatively protected compared to capital income.

5 Payments to Capital as a Share of GDP

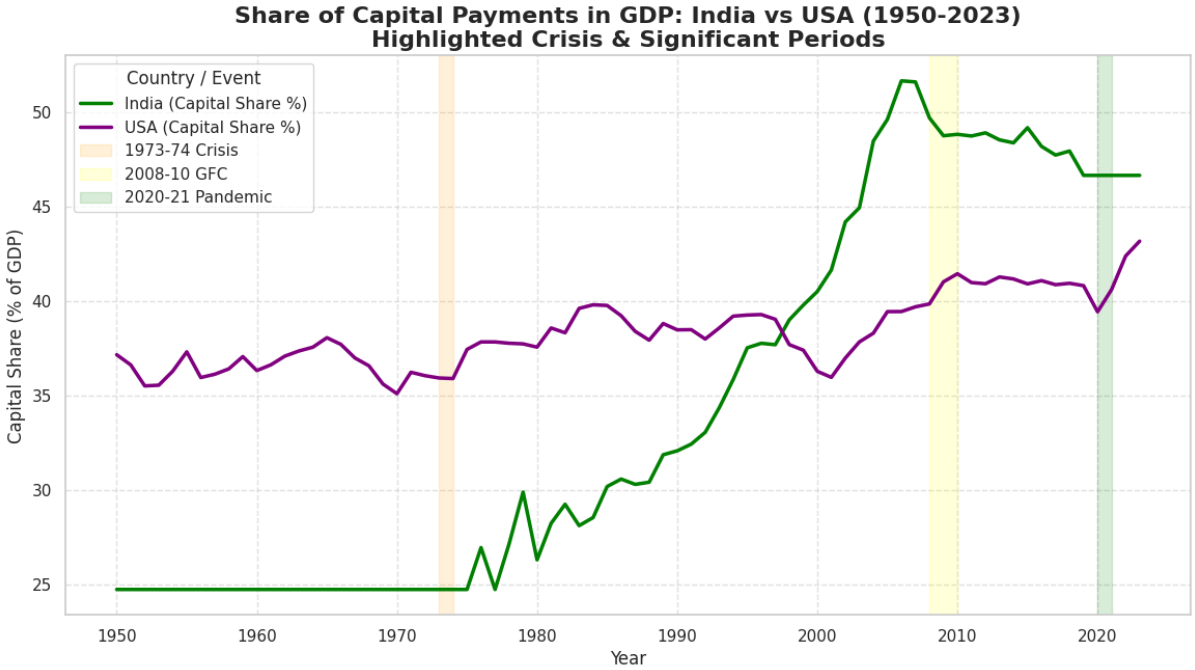

Figure 5: Capital Share of GDP: India vs USA

The share of capital payments in GDP was constant in India till 1970s and then increased sharply till late 2000s and then showing a slight decreasing trend. While the share in the US was quite stable till 2000 and then showing an increasing trend with a sharp increase in 2020.

During the 1973 oil crisis, the capital share remained largely stable in both countries, as both wages and profits adjusted simultaneously to higher production costs, leaving the overall distribution between labor and capital unchanged.

During the 2008 global financial crisis, there was a slowdown in global demand and a fall in capital inflows to India . This created uncertainty and firms reduced investment in India. This led to a fall in profits and hence return to capital. At the same time, wages and consumption did not fall as sharply, so the relative share of capital income in GDP declined in India. On the other hand in the USA, during the crisis, there were large job losses and wage cuts, so labour income fell sharply. In contrast, although profits declined, they did not fall as much as wages, and firms were able to protect margins through cost-cutting and productivity adjustments. As a result, the share of labour in GDP decreased more, which mechanically led to an increase in the capital share.

In 2020 (COVID period), the capital share in GDP remained relatively constant in India but increased in the US due to differences in how labour and profits were affected. In India, both profits and wages declined together during the lockdown due to a broad

economic contraction, especially in the informal sector. Since both components fell proportionately, the capital share remained relatively stable. In the US, labour income fell more sharply due to large-scale job losses, especially in contact-intensive sectors. At the same time, many large firms (especially in tech and capital-intensive sectors) maintained or even increased profits. As a result, the labour share declined, leading to an increase in capital share.

6 Growth Rate of Output per Capita

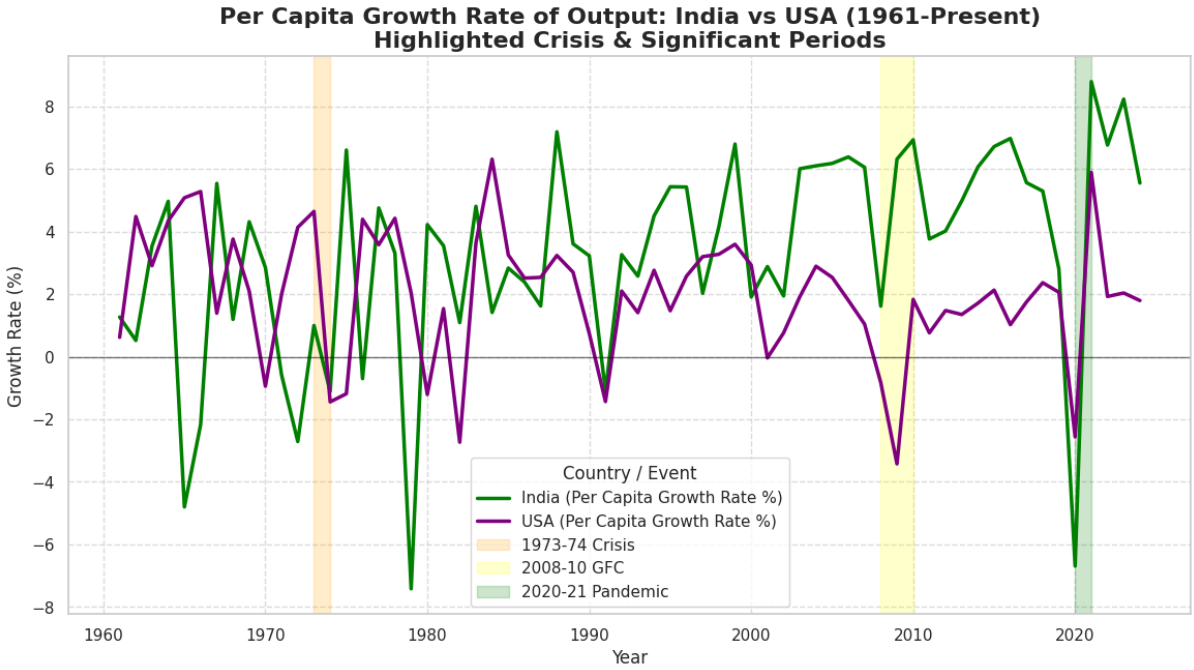

Figure 6: Per Capita Output Growth: India vs USA

Both India and the US show frequent fluctuations in per capita growth, indicating that economic growth is inherently cyclical in both countries. There are multiple ups and downs rather than a smooth trend.

However, the magnitude differs. India experiences larger swings (sharp peaks and deep drops), while the US shows smaller, more contained fluctuations. So, volatility exists in both, but it is more intense in India and more moderate in the US.

The graph also highlights that both economies experience synchronized downturns during major global shocks, but the magnitude differs. India tends to experience larger swings, both during downturns and recoveries, while the United States shows more moderate but persistent cycles. Overall, per capita output growth is highly cyclical, with India characterized by high-growth-high-volatility dynamics and the United States by low-growth-low-volatility stability.

In 2020, the increase (or sharp movement) appears stronger for India mainly because of the sudden and strict lockdown shock. India experienced an abrupt halt in economic activity, leading to a very sharp decline and immediate fluctuations within the year. Since the economy is more sensitive and less cushioned, even small reopening phases or sectoral recoveries show up as sharp movements in growth. In contrast, the US had large fiscal stimulus and support measures, which smoothed the fall in economic activity. As a result, changes in growth were less abrupt and more gradual.

7 Growth Rate of Consumption per Capita

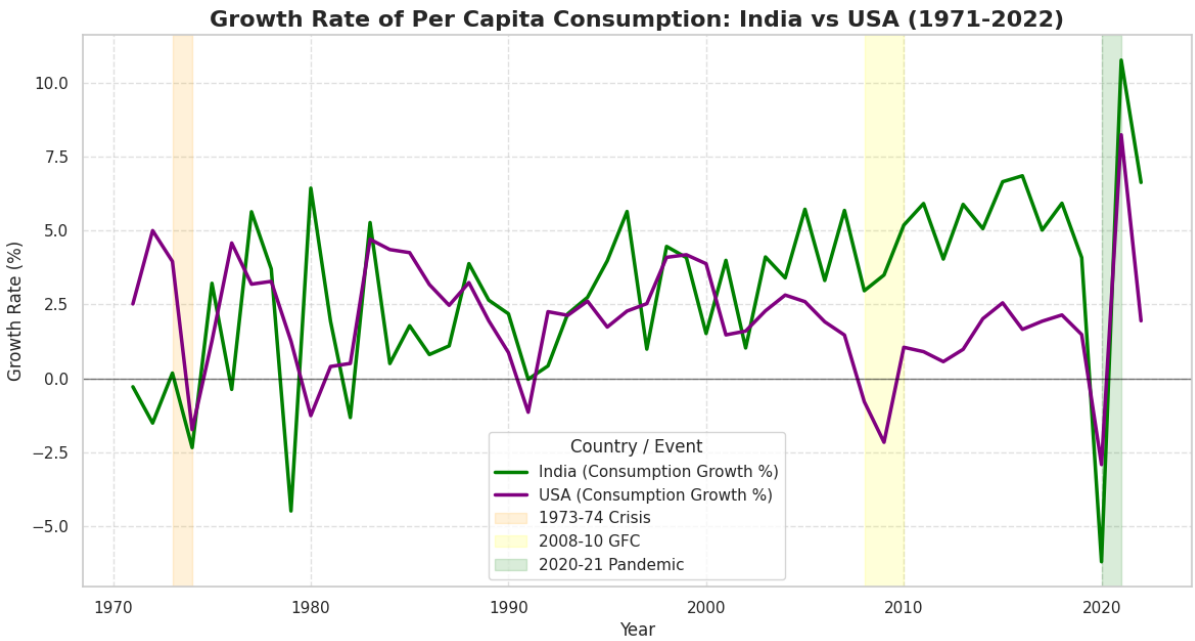

Figure 7: Per Capita Consumption Growth: India vs USA

India's per capita output growth (green) shows large fluctuations, with sharp increases and deep declines, often moving well beyond ±5%. In contrast, the USA's growth (yellow) stays within a narrower band, typically between -5% and +5%, indicating relatively lower volatility. While both countries exhibit cyclical movements, India's amplitude is much higher, whereas the US shows more moderate swings.

In 2020 (COVID period), both countries experienced a sharp decline, but India's contraction was significantly deeper. This was followed by a much sharper rebound in India, mainly due to the base effect and reopening of economic activity. The US also recovered, but the increase was more gradual due to a smaller initial decline and smoother policy support. Earlier periods also show India's growth becoming more consistent after the 1991 liberalisation, while the US maintains relatively stable cycles throughout.

The higher volatility in India reflects a developing economy sensitive to shocks, policy changes, and structural transformation, while the US reflects a mature economy with built-in stabilisers. Despite volatility, India's higher growth potential indicates convergence toward developed economies. The graph highlights the contrast between high-growth, high-volatility (India) and stable, moderate-growth (US) dynamics.

8 Unemployment rate

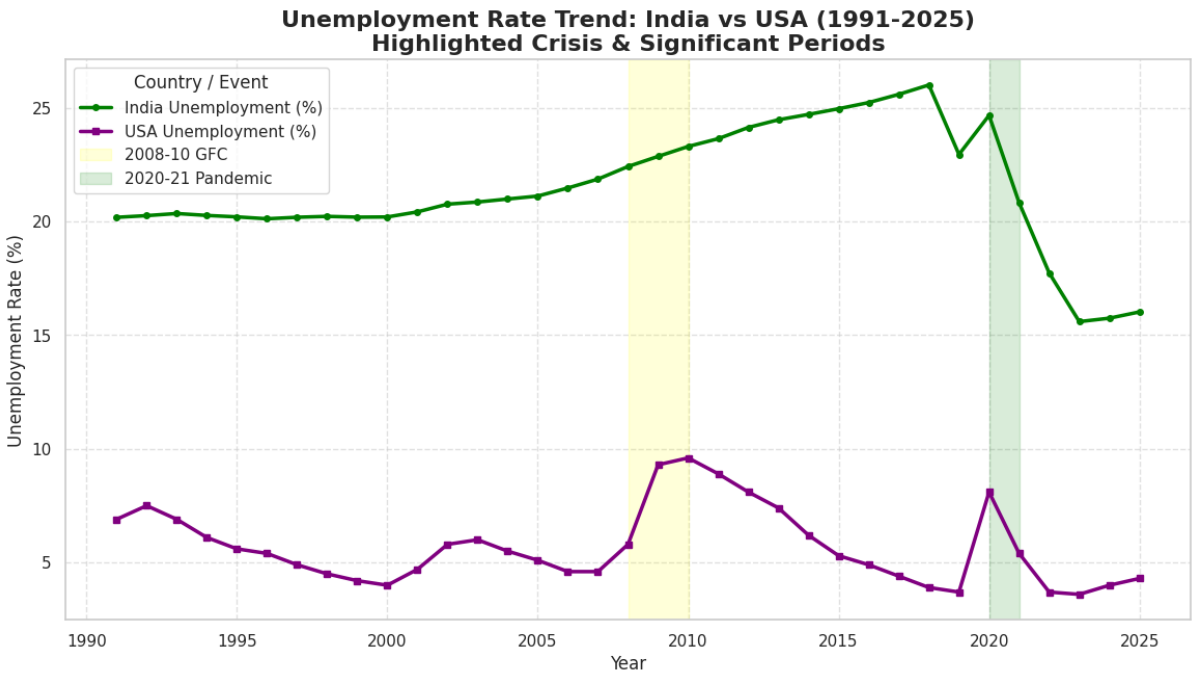

Figure 8: Unemployment rate: India vs USA

India's unemployment rate shows a gradual upward trend from 2000 to around 2018, followed by a decline after 2020. In contrast, the US shows clear cyclical movements, rising during crises and falling during recoveries. Overall, India's unemployment remains consistently higher, while the US fluctuates within a lower range.

The US exhibits higher short-term volatility, with sharp spikes (around 2008 and 2020) followed by quick declines. India, on the other hand, shows more gradual and persistent changes, indicating structural rather than cyclical unemployment. However, India does show a sharp drop after 2020, indicating some recent volatility.

During the 2008 financial crisis, the US experienced a sharp spike in unemployment, reflecting the severe impact of the crisis on jobs, while India showed only a gradual increase.

In the COVID-19 pandemic, both countries saw a significant rise in unemployment; however, the increase was much sharper in the US, followed by a quick recovery.

9 Average Duration of Unemployment

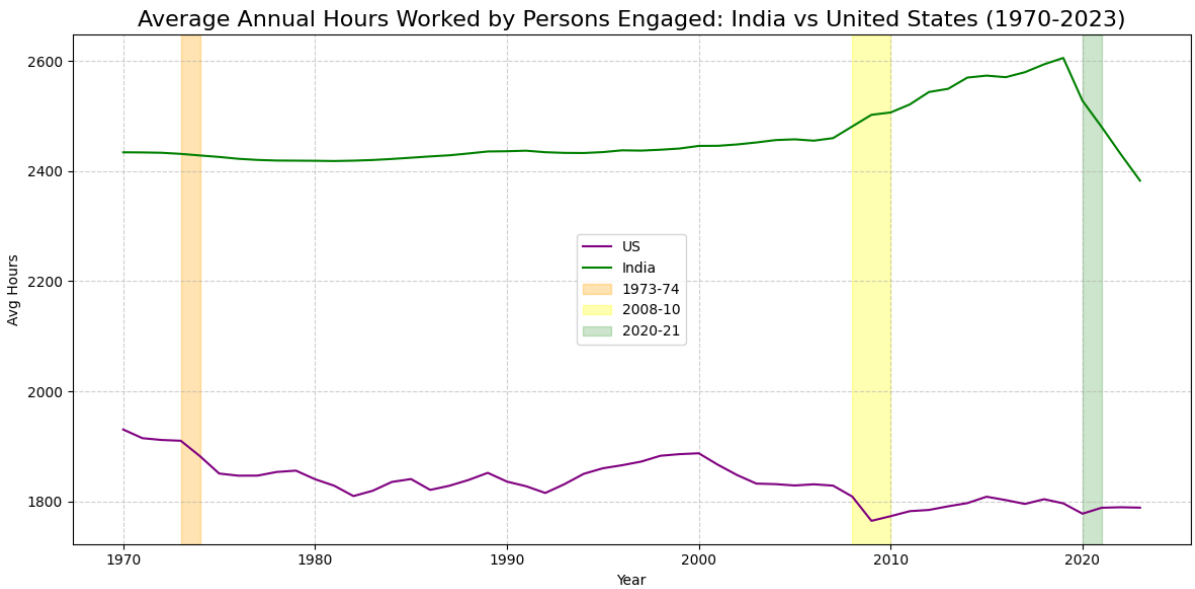

Figure 9: Average Annual Hours Worked: India vs USA

India's average annual hours worked are consistently much higher than those of the US and show a gradual upward trend until recent years, followed by a decline after 2020. In contrast, the US shows a long-term declining trend, with average hours falling from around 1900+ hours in the 1970s to around 1750–1800 hours in recent years. Overall, India reflects longer working hours, while the US reflects reduced working time over time.

Both countries show relatively low volatility, with smooth and gradual changes rather than sharp fluctuations. However, India exhibits slightly more variation in recent years.

During the 1973 oil crisis, the graph shows a decline in average hours worked in the US, while India remains relatively stable with only a slight dip.In the US, the oil shock led to an economic slowdown, reduced production, and lower labour demand. Firms cut back working hours due to higher costs and uncertainty, resulting in a noticeable decline in average hours worked. In India, the effect is much more muted. The economy was less industrialised and more dependent on agriculture and informal employment, where working hours are less responsive to short-term shocks. As a result, average hours remain relatively stable.

During the 2008 financial crisis, the graph shows a clear decline in average hours worked in the US, while India continues to rise initially and then stabilises. In the US, the financial crisis led to a sharp slowdown, job losses, and reduced working hours. Firms cut labour utilization, resulting in a noticeable drop in average hours worked. In India, the impact was milder. The economy did not contract as sharply, and employment remained

more stable due to the large informal sector. As a result, average hours worked did not fall significantly and even continued to increase slightly before stabilising.

During the COVID-19 pandemic, the graph shows a sharp decline in average hours worked in both India and the US, with a much steeper drop in India. In India, strict lockdowns caused a sudden halt in economic activity, especially in informal and labour-intensive sectors. This led to a large reduction in working hours, reflected as a sharp fall in the graph. In the US, there was also a decline due to business closures and reduced activity. However, strong policy support (stimulus, job protection measures) helped cushion the impact, so the fall in working hours was smaller and more gradual.

10 Inflation

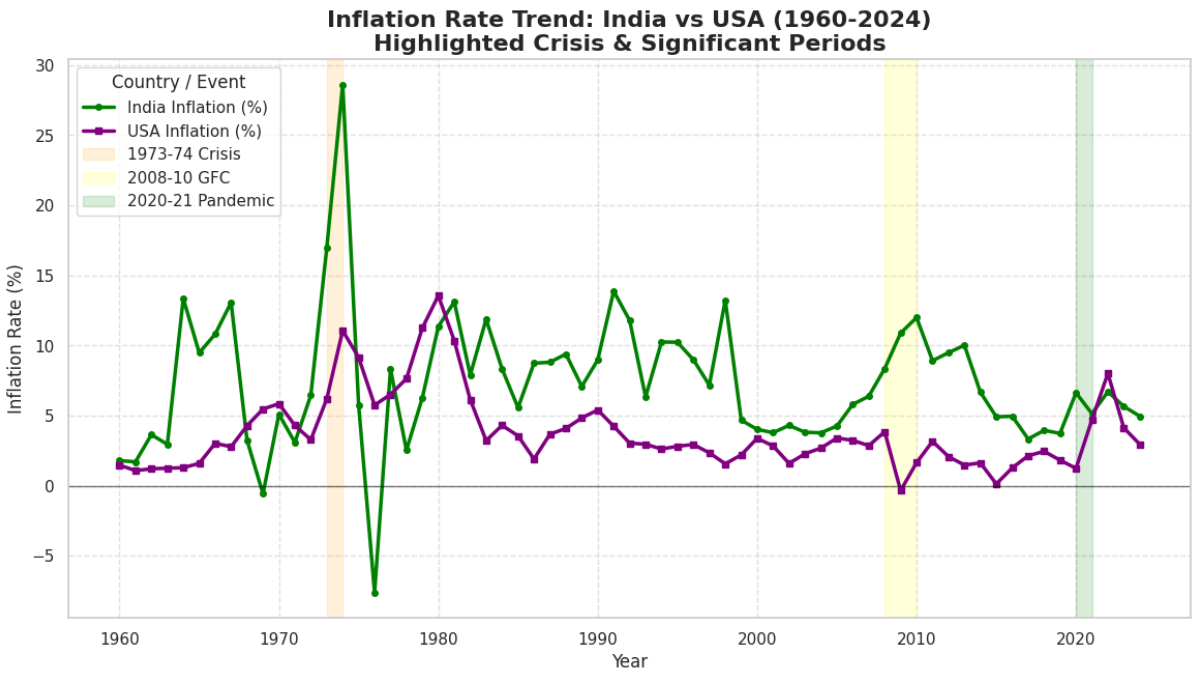

Figure 10: Inflation Rate: India vs USA

Inflation shows clear differences between India and the United States in both level and volatility. India exhibits higher and more volatile inflation, with frequent spikes driven by supply shocks, food prices, and structural factors. In contrast, the United States shows lower and more stable inflation, reflecting stronger monetary policy credibility and anchored expectations.

During the 1973 oil crisis, inflation rose sharply in both countries due to a surge in global oil prices, but the increase was much more pronounced in India due to higher dependence on imports and weaker stabilization mechanisms. The increase was very high for india because of its very high dependency on imported oil. It was not much higher for US because it had stronger institutions and more diversified energy use.

In India, inflation rose because of high global commodity and oil prices in early 2008, along with strong domestic demand. Since India is more dependent on imports and has supply-side constraints, these price increases passed through quickly, leading to higher inflation.In the US, inflation rose initially in 2008 due to oil prices but fell sharply later in the year due to the financial crisis and collapse in demand. Overall, the dominant trend by the end of 2008 was a decline in inflation.

A Python Code for Graphs

A.1 Consumption as a Fraction of GDP

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
file_path = 'Untitled spreadsheet - Sheet1.csv'
df = pd.read_csv(file_path)

# 2. Clean column names (remove leading/trailing spaces)
df.columns = df.columns.str.strip()

# 3. Select only the first 4 columns as requested
df = df.iloc[:, :4]

# 4. Extract year for alignment
# India dates are in "YYYY-YY" format (e.g., 1950-51) -> extract 1950
df['india_year'] = df['dates'].str.split('-').str[0].astype(int)

# USA dates are in "YYYY-MM-DD" format -> extract year
df['usa_year'] = pd.to_datetime(df['date USA']).dt.year

# 5. Create separate DataFrames for each country and merge to align by Year
df_india = df[['india_year', 'india perecent']].rename(columns={'india_year': 'Year', 'india perecent': 'India'})
df_usa = df[['usa_year', 'USA PERCent']].rename(columns={'usa_year': 'Year', 'USA PERCent': 'USA'})

# Merge on Year and sort chronologically
df_plot = pd.merge(df_india, df_usa, on='Year').sort_values('Year')

# 6. Plotting with Seaborn and Matplotlib
sns.set_theme(style="whitegrid")
plt.figure(figsize=(14, 7))
```

```python

# Plot line charts
sns.lineplot(data=df_plot, x='Year', y='India', label='India (Consumption % of GDP)', linewidth=2.5, color='green')
sns.lineplot(data=df_plot, x='Year', y='USA', label='USA (Consumption % of GDP)', linewidth=2.5, color='purple')

# --- 7. Highlight Specific Years ---
# Use plt.axvspan(start, end, ...) to create shaded background regions

# Highlight 1973, 74 (Oil crisis era)
plt.axvspan(1973, 1974, color='orange', alpha=0.2, label='1973-74 Crisis')

# Highlight 2008 to 2010 (Global Financial Crisis)
plt.axvspan(2008, 2010, color='yellow', alpha=0.2, label='2008-10 GFC')

# Highlight 2020, 2021 (Covid-19 Pandemic)
plt.axvspan(2020, 2021, color='green', alpha=0.2, label='2020-21 Pandemic')

# 8. Formatting the chart
plt.title('Consumption Share in GDP: India vs USA (1950-2025)\nHighlighted Crisis & Significant Periods', fontsize=16)
plt.xlabel('Year', fontsize=12)
plt.ylabel('Consumption (% of GDP)', fontsize=12)
plt.legend(title='Country / Event', frameon=True, loc='upper right')
plt.grid(True, linestyle='--', alpha=0.7)

# 9. Save and Show
plt.savefig('highlighted_consumption_trend.png', dpi=300, bbox_inches='tight')
plt.show()

print("Processing complete. The chart now highlights 1973-74, 2008-10, and 2020-21.")
```

A.2 Investment as a Fraction of GDP

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
file_path = 'gfgc - Sheet1.csv'
df = pd.read_csv(file_path)

# 2. Clean column names (remove leading/trailing spaces)
df.columns = df.columns.str.strip()

# 3. Select only the first 4 columns as requested
df = df.iloc[:, :4]

# --- 4. Process Data for Alignment ---

# India: Extract start year from "2023-24" -> 2023 and clean numerical strings
df['Year_India'] = df['india time pd'].str.split('-').str[0].astype(int)
df['India_GFCF'] = df['GFCF'].str.replace(',', '').astype(float)

# USA: Extract year from date and keep numeric value
df['Year_USA'] = pd.to_datetime(df['observation_date']).dt.year
df['USA_GFCF'] = df['USAGFCFQDSNAQ'].astype(float)

# 5. Align both countries by Year
df_india = df[['Year_India', 'India_GFCF']].rename(columns={'Year_India': 'Year'})
df_usa = df[['Year_USA', 'USA_GFCF']].rename(columns={'Year_USA': 'Year'})
df_plot = pd.merge(df_india, df_usa, on='Year').sort_values('Year')

# 6. Plotting with Seaborn and Dual Y-Axis
sns.set_theme(style="whitegrid")
fig, ax1 = plt.subplots(figsize=(14, 7))

```

```python
# Plot India Data (Left Axis)
color_india = 'tab:blue'
ax1.set_xlabel('Year', fontsize=12)
ax1.set_ylabel('India GFCF (Million    )', color=color_india, fontsize=12)
ax1.plot(df_plot['Year'], df_plot['India_GFCF'], color="green", label='India GFCF', linewidth=2.5)
ax1.tick_params(axis='y', labelcolor=color_india)

# Plot USA Data (Right Axis)
ax2 = ax1.twinx()
color_usa = 'tab:red'
ax2.set_ylabel('USA GFCF (USD $)', color=color_usa, fontsize=12)
ax2.plot(df_plot['Year'], df_plot['USA_GFCF'], color="purple", label='USA GFCF', linewidth=2.5)
ax2.tick_params(axis='y', labelcolor=color_usa)

# --- 7. Highlight Specified Year Ranges ---

# 1973, 1974 (Oil Crisis Era)
ax2.axvspan(1973, 1974, color='orange', alpha=0.2, label='1973-74 Crisis')

# 2008 to 2010 (Global Financial Crisis)
ax2.axvspan(2008, 2010, color='yellow', alpha=0.2, label='2008-10 GFC')

# 2020, 2021 (Pandemic Impact)
ax2.axvspan(2020, 2021, color='green', alpha=0.2, label='2020-21 Pandemic')

# 8. Title and Legend adjustments
plt.title('Trend of Gross Fixed Capital Formation (GFCF): India vs USA (1972-2023)\nHighlighted Significant Eras', fontsize=16)

# Merging legends from both axes
lines_1, labels_1 = ax1.get_legend_handles_labels()
lines_2, labels_2 = ax2.get_legend_handles_labels()
```

```python
ax2.legend(lines_1 + lines_2, labels_1 + labels_2, loc='upper left', frameon=True, title='Metric / Event')

fig.tight_layout()
plt.grid(True, linestyle='--', alpha=0.5)

# 9. Save and Show
plt.savefig('gfcf_highlighted_trend.png', dpi=300, bbox_inches='tight')
plt.show()

# Save the ordered data to a new file
df_plot.to_csv('gfcf_ordered_aligned.csv', index=False)
```

A.3 Government Spending as a Fraction of GDP

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load and clean the dataset
df = pd.read_csv('govt cons - Sheet1.csv')

# India: Extract start year from "2022-23" and rename columns
df_india = df[['year_india', '% govt cons of gdp india']].copy()
df_india['Year'] = df_india['year_india'].str.split('-').str[0].astype(int)
df_india = df_india.rename(columns={'% govt cons of gdp india': 'India (% of GDP)'})

# USA: Use Year_usa and rename columns
df_usa = df[['Year_usa', '% govt cons of gdp USA']].copy()
df_usa = df_usa.rename(columns={'Year_usa': 'Year', '% govt cons of gdp USA': 'USA (% of GDP)'})

# 2. Merge on common Year and sort chronologically
df_final = pd.merge(df_india[['Year', 'India (% of GDP)']],
                    df_usa[['Year', 'USA (% of GDP)']],
                    on='Year', how='inner').sort_values('Year')
```

```python

# 3. Plotting with Seaborn and Matplotlib
sns.set_theme(style="whitegrid")
plt.figure(figsize=(14, 7))

# Trend lines
plt.plot(df_final['Year'], df_final['India (% of GDP)'], label='India (Govt. Cons. % of GDP)',
         color='green', linewidth=2.5, markersize=4)
plt.plot(df_final['Year'], df_final['USA (% of GDP)'], label='USA (Govt. Cons. % of GDP)',
         color='purple', linewidth=2.5, markersize=4)

# 4. Highlight specific crisis/significant periods
plt.axvspan(1973, 1974, color='orange', alpha=0.2, label='1973-74 Crisis')
plt.axvspan(2008, 2010, color='yellow', alpha=0.2, label='2008-10 GFC')
plt.axvspan(2020, 2021, color='green', alpha=0.2, label='2020-21 Pandemic')

# 5. Formatting the chart
plt.title('Trend of Government Consumption (% of GDP): India vs USA (1970-2022)', fontsize=16, fontweight='bold')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Government Consumption (% of GDP)', fontsize=12)
plt.legend(title='Country / Event', frameon=True, loc='upper right')
plt.grid(True, linestyle='--', alpha=0.6)

plt.savefig('govt_consumption_trend.png', dpi=300, bbox_inches='tight')
plt.show()

# Export aligned data
df_final.to_csv('ordered_govt_cons.csv', index=False)
```

A.4 Payments to Labor as a Share of GDP

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
file_name = 'payments to labour - Sheet1.csv'
df = pd.read_csv(file_name)
df.columns = df.columns.str.strip()

# 2. Extract Year from observation_date
# Most dates are in YYYY-MM-DD format
df['Year'] = pd.to_datetime(df['observation_date']).dt.year

# 3. Convert decimal shares to percentages (0.75 -> 75.0)
india_col = 'Share of Labour Compensation in GDP'
usa_col = 'share of lab compensation in US'

df['India (% of GDP)'] = df[india_col] * 100
df['USA (% of GDP)'] = df[usa_col] * 100

# 4. Sort data by Year
df = df.sort_values('Year')

# 5. Plotting with Seaborn and Matplotlib
sns.set_theme(style="whitegrid")
plt.figure(figsize=(14, 7))

# Plot trend lines for both countries
plt.plot(df['Year'], df['India (% of GDP)'], label='India (Labour Share %)',
         color='green', linewidth=2.5, markersize=3)
plt.plot(df['Year'], df['USA (% of GDP)'], label='USA (Labour Share %)',
         color='purple', linewidth=2.5, markersize=3)

# 6. Highlight Specified Year Ranges using axvspan
# 1973-74 (Oil Crisis)
plt.axvspan(1973, 1974, color='orange', alpha=0.15, label='1973-74 Crisis')
```

```python

# 2008-2010 (Global Financial Crisis)
plt.axvspan(2008, 2010, color='yellow', alpha=0.15, label='2008-10 GFC')

# 2020-2021 (Pandemic)
plt.axvspan(2020, 2021, color='green', alpha=0.15, label='2020-21 Pandemic')

# 7. Formatting the chart
plt.title('Share of Labour Compensation in GDP: India vs USA (1950-2023)\nHighlighted Crisis & Significant Periods', fontsize=16, fontweight='bold')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Labour Share (% of GDP)', fontsize=12)
plt.legend(title='Country / Event', frameon=True, loc='lower left')
plt.grid(True, linestyle='--', alpha=0.6)

# 8. Save and export the chart
plt.savefig('labour_share_trend.png', dpi=300, bbox_inches='tight')
plt.show()

# Export the properly ordered data
df.to_csv('ordered_labour_share.csv', index=False)
```

A.5 Payments to Capital as a Share of GDP

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
file_name = 'payments to labour - Sheet1.csv'
df = pd.read_csv(file_name)
df.columns = df.columns.str.strip()

# 2. Process Date and extract Year
```

```python
df['Year'] = pd.to_datetime(df['observation_date']).dt.year

# 3. Calculate Capital Share (1 - Labour Share) and convert to percentage
# India's lab share is in 'Share of Labour Compensation in GDP'
# USA's lab share is in 'share of lab compensation in US'
df['India Capital Share (%)'] = (1 - df['Share of Labour Compensation in GDP']) * 100
df['USA Capital Share (%)'] = (1 - df['share of lab compensation in US']) * 100

# 4. Sort data chronologically
df = df.sort_values('Year')

# 5. Plotting with Seaborn and Matplotlib
sns.set_theme(style="whitegrid")
plt.figure(figsize=(14, 7))

# Plot trend lines
plt.plot(df['Year'], df['India Capital Share (%)'], label='India (Capital Share %)',
         color='green', linewidth=2.5, markersize=3)
plt.plot(df['Year'], df['USA Capital Share (%)'], label='USA (Capital Share %)',
         color='purple', linewidth=2.5, markersize=3)

# 6. Highlight requested periods
# 1973-74 (Oil Crisis)
plt.axvspan(1973, 1974, color='orange', alpha=0.15, label='1973-74 Crisis')
# 2008-2010 (Global Financial Crisis)
plt.axvspan(2008, 2010, color='yellow', alpha=0.15, label='2008-10 GFC')
# 2020-2021 (Pandemic)
plt.axvspan(2020, 2021, color='green', alpha=0.15, label='2020-21 Pandemic')

# 7. Formatting the chart
```

```
plt.title('Share of Capital Payments in GDP: India vs USA (1950-2023)\nHighlighted Crisis & Significant Periods', fontsize=16, fontweight='bold')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Capital Share (% of GDP)', fontsize=12)
plt.legend(title='Country / Event', frameon=True, loc='upper left')
plt.grid(True, linestyle='--', alpha=0.6)

# 8. Save the chart
plt.savefig('capital_share_trend.png', dpi=300, bbox_inches='tight')
plt.show()

# Export the calculated data
df.to_csv('ordered_capital_share.csv', index=False)
```

A.6 Growth Rate of Output per Capita

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
file_path = 'per capita Y gr - Sheet1.csv'
df = pd.read_csv(file_path)

# 2. Clean column names
df.columns = df.columns.str.strip()

# 3. Sort data by year to ensure continuous trend lines
df = df.sort_values('year')

# 4. Plotting with Seaborn and Matplotlib
sns.set_theme(style="whitegrid")
plt.figure(figsize=(14, 7))

# Plot India per capita growth rate
```

```
sns.lineplot(data=df, x='year', y='india', label='India (Per Capita Growth Rate %)',
             linewidth=2.5, color='green')

# Plot USA per capita growth rate
sns.lineplot(data=df, x='year', y='usa', label='USA (Per Capita Growth Rate %)',
             linewidth=2.5,color='purple')

# 5. Highlight Specified Year Ranges using axvspan
# 1973-74 (Oil Crisis)
plt.axvspan(1973, 1974, color='orange', alpha=0.2, label='1973-74 Crisis')

# 2008 to 2010 (Global Financial Crisis)
plt.axvspan(2008, 2010, color='yellow', alpha=0.2, label='2008-10 GFC')

# 2020, 2021 (Pandemic Period)
plt.axvspan(2020, 2021, color='green', alpha=0.2, label='2020-21 Pandemic')

# 6. Formatting the chart
plt.title('Per Capita Growth Rate of Output: India vs USA (1961-Present)\nHighlighted Crisis & Significant Periods', fontsize=16, fontweight='bold')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Growth Rate (%)', fontsize=12)
plt.legend(title='Country / Event', frameon=True, loc='best')

# Add a horizontal line at y=0 for reference
plt.axhline(0, color='black', linewidth=1, linestyle='-', alpha=0.5)

plt.grid(True, linestyle='--', alpha=0.7)

# 7. Save and display
plt.savefig('per_capita_growth_trend.png', dpi=300, bbox_inches='tight')
```

```
plt.show()

# Export the ordered data to CSV
df.to_csv('ordered_per_capita_growth.csv', index=False)
```

A.7 Growth Rate of Consumption per Capita

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
df = pd.read_csv('_Growth rate of consumption per capita - Sheet1
    .csv')
df.columns = df.columns.str.strip()

# 2. Ensure data is sorted by year
df = df.sort_values('year')

# 3. Plotting with Seaborn and Matplotlib
sns.set_theme(style="whitegrid")
plt.figure(figsize=(14, 7))

# Trend lines
plt.plot(df['year'], df['India'], label='India (Consumption
    Growth %)',
         color='green', linewidth=2.5)
plt.plot(df['year'], df['United States'], label='USA (Consumption
     Growth %)',
         color='purple', linewidth=2.5)

# 4. Highlight significant periods using axvspan
# 1973-74 (Oil Crisis)
plt.axvspan(1973, 1974, color='orange', alpha=0.15, label='
    1973-74 Crisis')

# 2008 to 2010 (Global Financial Crisis)
plt.axvspan(2008, 2010, color='yellow', alpha=0.15, label='
    2008-10 GFC')
```

```python

# 2020-21 (Pandemic Period)
plt.axvspan(2020, 2021, color='green', alpha=0.15, label='2020-21 Pandemic')

# 5. Add a reference line at 0 for positive/negative growth
plt.axhline(0, color='black', linewidth=1, linestyle='-', alpha=0.5)

# 6. Formatting the chart
plt.title('Growth Rate of Per Capita Consumption: India vs USA (1971-2022)', fontsize=16, fontweight='bold')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Growth Rate (%)', fontsize=12)
plt.legend(title='Country / Event', frameon=True, loc='best')
plt.grid(True, linestyle='--', alpha=0.6)

# 7. Save the final chart
plt.savefig('consumption_pc_growth_trend.png', dpi=300, bbox_inches='tight')
plt.show()

# Export the properly ordered data
df.to_csv('ordered_consumption_pc_growth.csv', index=False)
```

A.8 Unemployment rate

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
file_name = 'India unemp rate - India unemp rate.csv'
df = pd.read_csv(file_name)

# 2. Clean column names and extract Year
df.columns = df.columns.str.strip()
df['Year'] = pd.to_datetime(df['observation_date']).dt.year

```

```python
# 3. Sort by Year to ensure continuous trend lines
df = df.sort_values('Year')

# 4. Plotting with Seaborn and Matplotlib
sns.set_theme(style="whitegrid")
plt.figure(figsize=(14, 7))

# Identify column names
india_col = 'india unemployment'
usa_col = 'us unemployment'

# Plot trend lines
plt.plot(df['Year'], df[india_col], label='India Unemployment (%)',
         color='green', linewidth=2.5, marker='o', markersize=4)
plt.plot(df['Year'], df[usa_col], label='USA Unemployment (%)',
         color='purple', linewidth=2.5, marker='s', markersize=4)

# 5. Highlight specified periods using axvspan
# 1973-74 Crisis (Occurs before this specific dataset starts)
plt.axvspan(1973, 1974, color='orange', alpha=0.15, label='1973-74 Crisis')

# 2008-10 Global Financial Crisis
plt.axvspan(2008, 2010, color='yellow', alpha=0.15, label='2008-10 GFC')

# 2020-21 Pandemic
plt.axvspan(2020, 2021, color='green', alpha=0.15, label='2020-21 Pandemic')

# 6. Formatting the chart
plt.title('Unemployment Rate Trend: India vs USA (1991-2025)\nHighlighted Crisis & Significant Periods', fontsize=16, fontweight='bold')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Unemployment Rate (%)', fontsize=12)
plt.legend(title='Country / Event', frameon=True, loc='best')
plt.grid(True, linestyle='--', alpha=0.6)
```

```python

# 7. Save and export
plt.savefig('unemployment_trend.png', dpi=300, bbox_inches='tight')
plt.show()

# Export the ordered data
df.to_csv('ordered_unemployment_data.csv', index=False)
```

A.9 Average Duration of Unemployment (Proxy: Hours Worked)

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load datasets (correct files)
us = pd.read_csv('/content/AVHWPEUSA065NRUG.csv')
india = pd.read_csv('/content/AVHWPEINA065NRUG.csv')

# Extract Year from observation_date
us['Year'] = pd.to_datetime(us['observation_date']).dt.year
india['Year'] = pd.to_datetime(india['observation_date']).dt.year

# Rename columns
us = us[['Year', 'AVHWPEUSA065NRUG']].rename(columns={'AVHWPEUSA065NRUG': 'US'})
india = india[['Year', 'AVHWPEINA065NRUG']].rename(columns={'AVHWPEINA065NRUG': 'India'})

# Merge datasets on Year
df = pd.merge(us, india, on='Year', how='inner')

# Convert Year to numeric
df['Year'] = pd.to_numeric(df['Year'])

# Plot
plt.figure(figsize=(12, 6))

plt.plot(df['Year'], df['US'], color='purple', label='US')
plt.plot(df['Year'], df['India'], color='green', label='India')
```

```python

# Highlight periods
plt.axvspan(1973, 1974, color='orange', alpha=0.3, label='1973-74')
plt.axvspan(2008, 2010, color='yellow', alpha=0.3, label='2008-10')
plt.axvspan(2020, 2021, color='green', alpha=0.2, label='2020-21')

# Labels and title
plt.xlabel('Year')
plt.ylabel('Avg Hours')
plt.title('Average Annual Hours Worked by Persons Engaged: India vs United States (1970-2023)',  fontsize=16)

# Grid
plt.grid(True, linestyle='--', alpha=0.6)

# Legend
plt.legend()

# Show plot
plt.tight_layout()
plt.show()
```

A.10 Inflation

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
df = pd.read_csv('inflation - Sheet1.csv')
df.columns = df.columns.str.strip()

# 2. Convert observation_date to Year and sort
df['Year'] = pd.to_datetime(df['observation_date']).dt.year
df = df.sort_values('Year')

```

```python
# 3. Plotting with Seaborn and Matplotlib
sns.set_theme(style="whitegrid")
plt.figure(figsize=(14, 7))

# Trend lines
plt.plot(df['Year'], df['india inflation'], label='India Inflation (%)',
         color='green', linewidth=2.5, marker='o', markersize=4)
plt.plot(df['Year'], df['inflatiion USA'], label='USA Inflation (%)',
         color='purple', linewidth=2.5, marker='s', markersize=4)

# 4. Highlight significant periods using axvspan
# 1973-74 (Oil Crisis)
plt.axvspan(1973, 1974, color='orange', alpha=0.15, label='1973-74 Crisis')

# 2008 to 2010 (Global Financial Crisis)
plt.axvspan(2008, 2010, color='yellow', alpha=0.15, label='2008-10 GFC')

# 2020-21 (Pandemic Period)
plt.axvspan(2020, 2021, color='green', alpha=0.15, label='2020-21 Pandemic')

# 5. Add a reference line at 0 for reference
plt.axhline(0, color='black', linewidth=1, linestyle='-', alpha=0.5)

# 6. Formatting the chart
plt.title('Inflation Rate Trend: India vs USA (1960-2024)\nHighlighted Crisis & Significant Periods', fontsize=16, fontweight='bold')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Inflation Rate (%)', fontsize=12)
plt.legend(title='Country / Event', frameon=True, loc='best')
plt.grid(True, linestyle='--', alpha=0.6)

# 7. Save and display
```

```
plt.savefig('inflation_trend.png', dpi=300, bbox_inches='tight')
plt.show()

# Export the properly ordered data
df.to_csv('ordered_inflation_data.csv', index=False)
```

B Data Source

B.1 Consumption as a Fraction of GDP

https://data.rbi.org.in/DBIE/#/dbie/reports/Statistics/Real%20Sector/National%20IncomeComponents of Gross Domestic Product (at Market Prices)

https://fred.stlouisfed.org/series/PCECTPI

B.2 Investment as a Fraction of GDP

https://data.rbi.org.in/DBIE/#/dbie/reports/Statistics/Real%20Sector/National%20IncomeComponents of Gross Domestic Product (at Market Prices)

https://fred.stlouisfed.org/series/A007RE1Q156NBEA

B.3 Government Spending as a Fraction of GDP

https://data.rbi.org.in/DBIE/#/dbie/reports/Statistics/Real%20Sector/National%20IncomeComponents of Gross Domestic Product (at Market Prices)

https://data.worldbank.org/indicator/NE.CON.GOVT.ZS

B.4 Payments to Labor as a Share of GDP

https://fred.stlouisfed.org/series/LABSHPINA156NRUG

https://fred.stlouisfed.org/series/LABSHPUSA156NRUG

B.5 Payments to Capital as a Share of GDP

https://fred.stlouisfed.org/series/LABSHPINA156NRUG

https://fred.stlouisfed.org/series/LABSHPUSA156NRUG

B.6 Growth Rate of Output per Capita

https://data.worldbank.org/indicator/NY.GDP.PCAP.KD.ZG

B.7 Growth Rate of Consumption per Capita

https://data.worldbank.org/indicator/NE.CON.PRVT.PC.KD.ZG

B.8 Unemployment rate

https://fred.stlouisfed.org/series/SLUEM1524ZSIND

https://fred.stlouisfed.org/series/UNRATE

B.9 Average Duration of Unemployment (Proxy: Avg. Hours Worked ler annum)

https://fred.stlouisfed.org/series/AVHWPEINA065NRUG

https://fred.stlouisfed.org/series/AVHWPEUSA065NRUG

B.10 Inflation

https://fred.stlouisfed.org/series/FPCPITOTLZGIND

https://fred.stlouisfed.org/series/FPCPITOTLZGUSA